

Arduino Based FRONT OFFICE VISITOR ALARM

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This Arduino-based visitor alarm is useful for offices, clinics and shops. It alerts you to the entry of a visitor by producing a beep sound every time the door is opened. The alarm automatically switches off after a few seconds.

Circuit and working

Circuit diagram of the Arduino-based visitor alarm is shown in Fig. 1. It is built around an Arduino Uno board (Board1), a reed switch (S1) and a piezobuzzer (PZ1).

Arduino Uno operates off a 9V DC power supply. It is based on ATmega328 microcontroller and has 14 digital input/output pins, six analogue input pins, a USB connection for programming the on-board microcontroller, a 9V-12V DC power jack, an ICSP header and reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller.

Reed switch S1 is connected to digital pin 2 of Arduino as the input. PZ1 and LED1 are connected to digital pins 7 and 8, respectively, as outputs. LED1 is connected through a 220-ohm resistor.

Initially, when the door is closed, the magnet on the door is close to the reed switch (normally-open type) on the door frame and hence the reed switch is closed. So pin 2 of Arduino is 'high'.

When the door is opened, the reed switch opens and pin 2 of Arduino becomes 'low' through a 10-kilo-ohm pull-down resistor. This turns on piezobuzzer PZ1 and LED1. After a few seconds, PZ1 and LED1 automatically turn off.

The software (office_visitor.ino) for this project is written in Arduino programming language. Arduino Uno is programmed using Arduino IDE

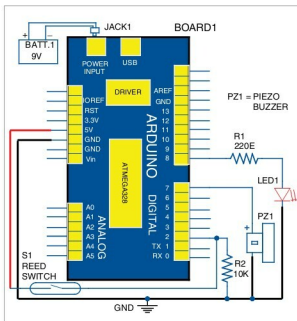


Fig. 1: Circuit diagram of Arduino-based office visitor alarm

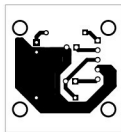


Fig. 2: Actual size PCB layout of Arduino-based office visitor alarm

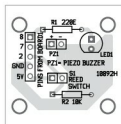


Fig. 3: Components layout for the PCB

software. ATmega328 on Arduino Uno comes with a bootloader that allows you to upload new code to it without using any external hardware programmer.

Construction and testing

An actual-size PCB layout of the Arduino-based front-office visitor alarm is shown in Fig. 2. After assembling the circuit, enclose it in a suitable cabinet. Affix the magnet on the door and the reed switch on the door frame such that when you close

PARTS LIST	
Semiconductors:	
Board1	- Arduino Uno
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 220-ohm
R2	- 10-kilo-ohm
Miscellaneous:	
PZ1	- Piezo buzzer
Jack1	- DC jack
BATT.1	- 9V PP3 battery
S1	- Reed switch (normally open type)

the door, the magnet comes closest possible to the reed switch. Use a 9V battery/adaptor to power the Arduino Uno board. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



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